$$
\begin{aligned}
&C_{u\phi_2}^{(11)\,i1\_i2\_} \to \frac{1}{16\,\pi^2}\Big(-\frac{1}{3}\, m_2\,\tilde{\mu}\, g_2{}^4 \left(y_u{}^{i1i2} - 2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{2,2,0}[m_2,\tilde{\mu}] +\\
&\frac{1}{6}\, m_2\,\tilde{\mu}\, g_2{}^4 \left(y_u{}^{i1i2} - 2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{3,2,-1}[m_2,\tilde{\mu}] -\\
&\frac{1}{4}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{pr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,2,0}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] - \frac{1}{4}\,\tilde{\mu}\, g_1{}^2\, y_e{}^{pr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{2,2,0}\left[m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^p\right] +\\
&\frac{1}{2}\,\tilde{\mu}\, g_2{}^2\, y_e{}^{pr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r\right] + \frac{1}{4}\,\tilde{\mu}\, g_1{}^2\, y_e{}^{pr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, y_e{}^{pr}\,\overline{a_e}{}^{pr}\left(y_u{}^{i1i2}\left(3\, g_1{}^2 - 11\, g_2{}^2\right) + 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{4,1,-1}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r\right] +\\
&\frac{1}{6}\,\tilde{\mu}\, y_e{}^{pr}\,\overline{a_e}{}^{pr}\left(y_u{}^{i1i2}\left(-3\, g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{5,1,-2}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r\right] + \frac{1}{2}\,\tilde{\mu}\, y_d{}^{pr}\, y_u{}^{i1i2}\\
&\quad \overline{a_d}{}^{pr}\left(-2\, g_1{}^2 + 3\, g_2{}^2\right) LF_{3,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r\right] + \frac{1}{4}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{pr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r\right] +\\
&\frac{3}{8}\,\tilde{\mu}\, y_d{}^{pr}\,\overline{a_d}{}^{pr}\left(y_u{}^{i1i2}\left(7\, g_1{}^2 - 11\, g_2{}^2\right) + 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{4,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r\right] +\\
&\frac{1}{2}\,\tilde{\mu}\, y_d{}^{pr}\,\overline{a_d}{}^{pr}\left(y_u{}^{i1i2}\left(-3\, g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{5,1,-2}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r\right] -\\
&\frac{1}{8}\,\tilde{\mu}\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2}\left(g_1{}^2 + 5\, g_2{}^2\right) - 4\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{2,2,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{3}{4}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] - \frac{1}{4}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2} + y_u{}^{si2}\,\delta_{si1}\right)\\
&\quad LF_{3,2,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] - \frac{3}{4}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{4}\,\tilde{\mu}\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2}\left(g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{3,1,0}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2}\left(g_1{}^2 + 5\, g_2{}^2\right) - 4\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{3,2,-1}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2}\left(g_1{}^2 - 4\, g_2{}^2\right) + 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{4,1,-1}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{2}\,\tilde{\mu}\, y_u{}^{pr}\,\overline{a_u}{}^{pr}\left(y_u{}^{i1i2}\left(-3\, g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{si2}\,\delta_{si1}\right) LF_{5,1,-2}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{4}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(-g_1{}^2 + g_2{}^2\right) LF_{3,1,0}[\tilde{\mu}, m_1] - \frac{1}{2}\, m_1\,\tilde{\mu}\, g_1{}^4\, y_u{}^{i1i2}\, LF_{3,2,-1}[\tilde{\mu}, m_1] +\\
&\frac{1}{8}\, m_1\,\tilde{\mu}\, g_1{}^2\left(y_u{}^{i1i2}\left(6\, g_1{}^2 - 7\, g_2{}^2\right) + 2\, g_2{}^2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{4,1,-1}[\tilde{\mu}, m_1] +\\
&\frac{1}{2}\, m_1\,\tilde{\mu}\, g_1{}^4\, y_u{}^{i1i2}\, LF_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{6}\, m_1\,\tilde{\mu}\, g_1{}^2\left(y_u{}^{i1i2}\left(-3\, g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{pi2}\,\delta_{pi1}\right)\\
&\quad LF_{5,1,-2}[\tilde{\mu}, m_1] + \frac{1}{12}\, m_2\,\tilde{\mu}\, g_2{}^2\left(y_u{}^{i1i2}\left(-9\, g_1{}^2 + 13\, g_2{}^2\right) - 8\, g_2{}^2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{3,1,0}[\tilde{\mu}, m_2] -\\
&\frac{1}{3}\, m_2\,\tilde{\mu}\, g_2{}^4\left(5\, y_u{}^{i1i2} + 2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{3,2,-1}[\tilde{\mu}, m_2] + \frac{1}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\\
&\quad \left(y_u{}^{i1i2}\left(18\, g_1{}^2 - 25\, g_2{}^2\right) + 14\, g_2{}^2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{4,1,-1}[\tilde{\mu}, m_2] + 2\, m_2\,\tilde{\mu}\, g_2{}^4\, y_u{}^{i1i2}\, LF_{4,2,-2}[\tilde{\mu}, m_2] +\\
&\frac{1}{2}\, m_2\,\tilde{\mu}\, g_2{}^2\left(y_u{}^{i1i2}\left(-3\, g_1{}^2 + 4\, g_2{}^2\right) - 2\, g_2{}^2\, y_u{}^{pi2}\,\delta_{pi1}\right) LF_{5,1,-2}[\tilde{\mu}, m_2] +\\
&\frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] + \frac{1}{72}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\\
&\quad LF_{2,2,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] - \frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{3,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{6}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, \tilde{\mu}\right] - \frac{1}{6}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\\
&\quad LF_{3,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, \tilde{\mu}\right] - \frac{1}{72}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{d}}{}^r\right] +\\
&\frac{1}{18}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,0}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] - \frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\\
&\quad LF_{2,2,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] - \frac{1}{18}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{3,1,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{72}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{2,2,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}\right] + \frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\\
&\quad y_u{}^{i1r}\, LF_{2,2,1,-1}\left[m_1, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^{i1}\right] - \frac{1}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,0}\left[m_1, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{3,1,1,-1}\left[m_1, m_{\tilde{u}}{}^r, \tilde{\mu}\right] + \frac{1}{72}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right)\\
&\quad LF_{2,2,1,-1}\left[m_1, m_{\tilde{u}}{}^{i2}, m_{\tilde{q}}{}^{i1}\right] - \frac{5}{24}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{d}}{}^r\right] +\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\left(-2\, y_u{}^{pi2}\left(\overline{y_d}{}^{pr}\, y_d{}^{i1r} + \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) + y_u{}^{i1i2}\left(-9\, g_1{}^2 + g_2{}^2\right)\right) LF_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{7}{24}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{12}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(9\, g_1{}^2 - g_2{}^2\right) LF_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}{}^{i2}\right] - \frac{3}{4}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\\
&\quad (3\, m_1 + m_2)\, LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + \tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(m_1 + m_2)\, LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] +\\
&\frac{1}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{d}}{}^r\right] - \frac{1}{2}\, m_2\,\tilde{\mu}\, g_2{}^4\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}{}^p\right]\delta_{pi1} +\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\left(6\, y_u{}^{pi2}\left(-\overline{y_d}{}^{pr}\, y_d{}^{i1r} + \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) + y_u{}^{i1i2}\left(3\, g_1{}^2 + 29\, g_2{}^2\right)\right) LF_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{1}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{u}}{}^r\right] -\\
&\frac{2}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,0}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] + \frac{2}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\\
&\quad LF_{3,1,1,-1}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] + \frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,2,1,-1}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{d}}{}^r\right] +\\
&\frac{2}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,0}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] - \frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\\
&\quad LF_{2,2,1,-1}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] - \frac{2}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{3,1,1,-1}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{6}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}\right] + \frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\\
&\quad LF_{2,2,1,-1}\left[m_3, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^{i1}\right] + \frac{1}{6}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{2,2,1,-1}\left[m_3, m_{\tilde{u}}{}^{i2}, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{24}\,\tilde{\mu}\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\left(-7\, g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, y_d{}^{i1r}\,\overline{a_d}{}^{pr}\left(y_u{}^{pi2}\left(g_1{}^2 - g_2{}^2\right) - 2\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{si2}\right) LF_{2,2,1,-1}\left[m_{\tilde{d}}{}^r, \tilde{\mu}, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{24}\,\tilde{\mu}\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\left(-5\, g_1{}^2 + 9\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, \tilde{\mu}\right] -\\
&3\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{u}}{}^r\right] +\\
&3\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{u}}{}^r\right] +\\
&\frac{9}{4}\,\tilde{\mu}\, y_d{}^{sr}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r\right] -\\
&\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] +\\
&\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] +\\
&3\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{d}}{}^t\right] +\\
&\frac{3}{8}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^s\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, y_d{}^{i1r}\,\overline{a_d}{}^{pr}\left(y_u{}^{pi2}\left(g_1{}^2 - g_2{}^2\right) + 2\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{si2}\right) LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^p, \tilde{\mu}, m_{\tilde{d}}{}^r\right] -\\
&\frac{1}{8}\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{si2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^p, \tilde{\mu}, m_{\tilde{u}}{}^r\right] -\\
&3\,\tilde{\mu}\, y_d{}^{sr}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,0}\left[m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&3\,\tilde{\mu}\, y_d{}^{sr}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,0}\left[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{3}{4}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{3}{8}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{108}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(5\, g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{u}}{}^{i2}\right] +\\
&\frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{u}}{}^{i2}\right] -\\
&\frac{1}{72}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1}, m_1, \tilde{\mu}\right] +\\
&\frac{1}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1}, m_2, \tilde{\mu}\right] -\\
&\frac{1}{9}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(5\, g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{q}}{}^{i1}, m_3, m_{\tilde{u}}{}^{i2}\right] +\\
&\frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_3, m_{\tilde{u}}{}^{i2}\right] -\\
&3\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,0}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t\right] +\\
&3\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,-1}\left[m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t\right] +\\
&\frac{3}{2}\,\tilde{\mu}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{8}\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{si2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,-1}\left[m_{\tilde{u}}{}^r, \tilde{\mu}, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{108}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(5\, g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{u}}{}^{i2}, m_1, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{36}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i2}, m_1, m_{\tilde{q}}{}^{i1}\right] - \frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^4\, y_u{}^{i1i2}\\
&\quad LF_{2,1,1,0}\left[m_{\tilde{u}}{}^{i2}, m_1, \tilde{\mu}\right] + \frac{1}{9}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(5\, g_1{}^2 + 3\, g_2{}^2\right) LF_{2,1,1,0}\left[m_{\tilde{u}}{}^{i2}, m_3, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1i2}\left(g_1{}^2 - g_2{}^2\right) LF_{3,1,1,-1}\left[m_{\tilde{u}}{}^{i2}, m_3, m_{\tilde{q}}{}^{i1}\right] + \frac{3}{2}\, m_1\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\\
&\quad LF_{2,1,1,0}[\tilde{\mu}, m_1, m_2] + \frac{1}{8}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(-31\, m_1 + 5\, m_2)\, LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] +\\
&\frac{3}{4}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(3\, m_1 - m_2)\, LF_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] + \frac{1}{2}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\\
&\quad LF_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{d}}{}^r\right] - \frac{1}{4}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{d}}{}^r\right] +\\
&\frac{1}{12}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\left(\overline{y_d}{}^{pr}\, y_d{}^{i1r} + \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) LF_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{1}{12}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\left(\overline{y_d}{}^{pr}\, y_d{}^{i1r} + \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) LF_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{1}{2}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{4}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{u}}{}^r\right] +\\
&2\, m_1\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\, LF_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] - \frac{1}{2}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{d}}{}^r\right] +\\
&\frac{1}{4}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}{}^r\right] +\\
&\frac{3}{4}\, m_2\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\left(\overline{y_d}{}^{pr}\, y_d{}^{i1r} - \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) LF_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{3}{4}\, m_2\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\left(-\overline{y_d}{}^{pr}\, y_d{}^{i1r} + \overline{y_u}{}^{pr}\, y_u{}^{i1r}\right) LF_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{2}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{u}}{}^r\right] - \frac{1}{4}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\\
&\quad LF_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{u}}{}^r\right] + \frac{1}{4}\,\tilde{\mu}\, y_d{}^{i1r}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{si2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{4}\,\tilde{\mu}\, y_d{}^{i1r}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{si2}\,\overline{a_d}{}^{pr}\, LF_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] - \frac{1}{4}\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{si2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\\
&\quad LF_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] + \frac{1}{4}\,\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{si2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{3}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(m_1 + m_2)\, LF_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] - \frac{1}{6}\, m_2\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\\
&\quad LF_{2,1,1,1,-1}\left[m_1, m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] - \frac{1}{8}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] - \frac{1}{36}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\\
&\quad LF_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] + \frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] +\\
&\frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}\right] +\\
&\frac{1}{8}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] -\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{2}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{u}}{}^{i2}, \tilde{\mu}\right] -\\
&\frac{1}{8}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{36}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{8}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r\right] -\\
&\frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^4\, y_u{}^{i1i2}\, LF_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}, \tilde{\mu}\right] +\\
&\frac{2}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_1, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{6}\, m_1\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\, LF_{2,1,1,1,-1}\left[m_2, m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{3}{8}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] -\\
&\frac{3}{16}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{4}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] -\\
&\frac{3}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] -\\
&\frac{3}{8}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{3}{16}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{4}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{3}{8}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{4}{3}\,\tilde{\mu}\, g_3{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,0}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] +\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p\right] +\\
&\frac{8}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,0}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}\right] +\\
&\frac{4}{3}\,\tilde{\mu}\, g_3{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{8}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{1,1,1,1,0}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{48}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{d}}{}^r, m_1, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_{\tilde{d}}{}^r, m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] -\\
&\frac{1}{16}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{d}}{}^r, m_2, m_{\tilde{q}}{}^p, \tilde{\mu}\right] -\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_{\tilde{d}}{}^r, m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{3}{4}\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s\right] -\\
&\tilde{\mu}\, y_d{}^{i1r}\,\overline{y_u}{}^{st}\, y_u{}^{pt}\, y_u{}^{si2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,0}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, \tilde{\mu}\right] -\\
&\frac{3}{4}\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^s\right] -\\
&\tilde{\mu}\,\overline{y_d}{}^{st}\, y_d{}^{i1t}\, y_u{}^{pi2}\, y_u{}^{sr}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,0}\left[m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{4}\,\tilde{\mu}^3\,\overline{y_e}{}^{st}\, y_e{}^{pt}\, y_e{}^{sr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s\right] -\\
&\frac{1}{4}\,\tilde{\mu}^3\,\overline{y_e}{}^{st}\, y_e{}^{pt}\, y_e{}^{sr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{l}}{}^s\right] -\\
&\frac{1}{2}\,\tilde{\mu}^3\,\overline{y_e}{}^{st}\, y_e{}^{pt}\, y_e{}^{sr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{2,1,1,1,0}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s\right] +\\
&\frac{1}{2}\,\tilde{\mu}^3\,\overline{y_e}{}^{st}\, y_e{}^{pt}\, y_e{}^{sr}\, y_u{}^{i1i2}\,\overline{a_e}{}^{pr}\, LF_{3,1,1,1,-1}\left[m_{\tilde{l}}{}^p, m_{\tilde{e}}{}^r, m_{\tilde{e}}{}^t, m_{\tilde{l}}{}^s\right] -\\
&\frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{1}{48}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_1, m_{\tilde{d}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{48}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_1, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_2, m_{\tilde{d}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{16}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_2, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] -\\
&\frac{3}{2}\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s\right] +\\
&\frac{3}{2}\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\, LF_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s\right] +\\
&3\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] -\\
&3\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] +\\
&\frac{9}{4}\,\tilde{\mu}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{u}}{}^t\right] -\\
&\frac{3}{4}\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^t, m_{\tilde{u}}{}^r\right] -\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t\right] +\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t\right] -\\
&3\,\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{pt}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^s\right] +\\
&\frac{3}{8}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t\right] +\\
&\frac{9}{8}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] -\\
&3\,\tilde{\mu}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\left[m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t\right] +\\
&3\,\tilde{\mu}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t\right] -\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\left[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t\right] +\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t\right] +\\
&\frac{3}{8}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^t\right] +\\
&3\,\tilde{\mu}\, y_d{}^{sr}\, y_u{}^{i1i2}\,\overline{a_d}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{3}{8}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{d}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{72}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_1, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_2, m_{\tilde{d}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{8}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_2, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_2, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{2}{3}\,\tilde{\mu}\, g_3{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{q}}{}^{i1}, m_3, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{48}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{u}}{}^r, m_1, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{48}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_{\tilde{u}}{}^r, m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{1}{16}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[m_{\tilde{u}}{}^r, m_2, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[m_{\tilde{u}}{}^r, m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}\right] +\\
&\frac{3}{4}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,2,1,1,-1}\left[m_{\tilde{u}}{}^r, m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s\right] -\\
&\frac{3}{2}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{2,1,1,1,0}\left[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] +\\
&\frac{3}{2}\,\tilde{\mu}\, y_u{}^{sr}\, y_u{}^{i1i2}\,\overline{a_u}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{3,1,1,1,-1}\left[m_{\tilde{u}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r\right] +\\
&\frac{1}{12}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(-7\, m_1 + 9\, m_2)\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{q}}{}^{i1}\right] +\\
&\frac{1}{3}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(3\, m_1 - m_2)\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_2, m_{\tilde{u}}{}^{i2}\right] -\\
&\frac{23}{48}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{2}\, m_1\, g_1{}^2\,\tilde{\mu}^3\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] -\\
&\frac{1}{16}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{23}{48}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{16}\, m_1\,\tilde{\mu}\, g_1{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] -\\
&\frac{9}{16}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{1}{2}\, m_2\, g_2{}^2\,\tilde{\mu}^3\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p\right] +\\
&\frac{3}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}\right] -\\
&\frac{7}{16}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r\right] -\\
&\frac{3}{16}\, m_2\,\tilde{\mu}\, g_2{}^2\,\overline{y_u}{}^{pr}\, y_u{}^{pi2}\, y_u{}^{i1r}\, LF_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{3}\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, y_u{}^{i1i2}\,(m_1 + m_2)\, LF_{1,1,1,1,1,-1}\left[m_1, m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}, \tilde{\mu}\right] -\\
&\frac{4}{3}\, m_1\, m_2\,\tilde{\mu}\, g_1{}^2\, g_2{}^2\, a_u{}^{i1i2}\, LF_{1,1,1,1,1,0}\left[m_1, m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{u}}{}^{i2}, \tilde{\mu}\right] +\\
&\frac{2}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\left(\overline{a_d}{}^{pr}\, a_u{}^{pi2} - \tilde{\mu}^2\,\overline{y_d}{}^{pr}\, y_u{}^{pi2}\right) LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}, \tilde{\mu}\right] +\\
&\frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\left(\overline{a_d}{}^{pr}\, a_u{}^{pi2} - \tilde{\mu}^2\,\overline{y_d}{}^{pr}\, y_u{}^{pi2}\right) LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}\right] -\\
&\frac{1}{6}\,\tilde{\mu}\, g_1{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,1,-1}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, \tilde{\mu}\right] -\\
&\frac{1}{6}\, m_1\, g_1{}^2\,\tilde{\mu}^3\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{2}{3}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, a_u{}^{pi2}\, LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{1}{6}\,\tilde{\mu}\, g_1{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,1,-1}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] -\\
&\frac{2}{9}\, m_1\,\tilde{\mu}\, g_1{}^2\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, a_u{}^{pi2}\, LF_{1,1,1,1,1,0}\left[m_1, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r\right] -\\
&\frac{1}{2}\,\tilde{\mu}\, g_2{}^2\, y_d{}^{i1r}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,1,-1}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{2}\, m_2\, g_2{}^2\,\tilde{\mu}^3\,\overline{y_d}{}^{pr}\, y_d{}^{i1r}\, y_u{}^{pi2}\, LF_{1,1,1,1,1,0}\left[m_2, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, \tilde{\mu}\right] +\\
&\frac{1}{2}\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,1,-1}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&m_2\,\tilde{\mu}\, g_2{}^2\, y_u{}^{pi2}\,\overline{a_u}{}^{pr}\, a_u{}^{i1r}\, LF_{1,1,1,1,1,0}\left[m_2, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^r, \tilde{\mu}\right] +\\
&\frac{8}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_d{}^{i1r}\left(\overline{a_d}{}^{pr}\, a_u{}^{pi2} - \tilde{\mu}^2\,\overline{y_d}{}^{pr}\, y_u{}^{pi2}\right) LF_{1,1,1,1,1,0}\left[m_3, m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}\right] -\\
&\frac{8}{3}\, m_3\,\tilde{\mu}\, g_3{}^2\, y_u{}^{i1r}\,\overline{a_u}{}^{pr}\, a_u{}^{pi2}\, LF_{1,1,1,1,1,0}\left[m_3, m_{\tilde{q}}{}^{i1}, m_{\tilde{q}}{}^p, m_{\tilde{u}}{}^{i2}, m_{\tilde{u}}{}^r\right] -\\
&\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{sr}\, y_d{}^{i1t}\, y_u{}^{pi2}\,\overline{a_d}{}^{pr}\, LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}{}^r, m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, \tilde{\mu}\right] -\\
&\tilde{\mu}\, y_d{}^{i1r}\, y_u{}^{si2}\,\overline{a_d}{}^{pr}\,\overline{a_u}{}^{st}\, a_u{}^{pt}\, LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}{}^r, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^t, \tilde{\mu}\right] +\\
&\tilde{\mu}^3\,\overline{y_d}{}^{st}\, y_d{}^{i1t}\, y_u{}^{pi2}\, y_u{}^{sr}\,\overline{a_u}{}^{pr}\, LF_{1,1,1,1,1,0}\left[m_{\tilde{d}}{}^t, m_{\tilde{q}}{}^p, m_{\tilde{q}}{}^s, m_{\tilde{u}}{}^r, \tilde{\mu}\right]\Big)
\end{aligned}
$$